

Aigerim Toleukhanova

Sherman Oaks, CA

+1 (310) 456- 4350

aigerata3@g.ucla.edu

www.linkedin.com/in/aigerimtoleukhanova

Objective

To secure a Database Administrator (DBA) or Data Analyst position where I can leverage my expertise in SQL database management, data analytics, and machine learning to optimize data processing, enhance system performance, and drive data-driven decision-making. I aim to apply my strong analytical skills, experience in database security, and knowledge of predictive modeling to contribute to enterprise data solutions and business intelligence initiatives.

TECHNICAL SUMMARY:

- Over 4 years of experience in database administration, including installation, configuration, management, monitoring, maintenance, performance optimization, and troubleshooting for MySQL, PostgreSQL, and SQL Server in research and project-based environments.
- 4 years of SQL experience, writing and debugging complex queries, stored procedures, functions, and performance tuning for optimized data retrieval.
- 3 years of experience in machine learning and predictive modeling, working with Bayesian Networks, Directed Acyclic Graphs, Random Forest, XGBoost, AdaBoost, GBM, SVM, and Naïve Bayes.
- 3 years of experience in data analytics and statistical modeling, including data mining, feature importance extraction, confusion matrix evaluation, and 10-fold cross-validation.
- 3+ years of experience in developing secure APIs and full-stack applications, integrating SQL databases, PHP backend, and Angular frontend for user authentication and data processing.
- 3+ years of experience in data security, including SQL injection prevention, role-based access control, database encryption, and secure user authentication.
- 2+ years of experience with big data processing and cloud-based SQL servers, including data warehousing, ETL pipelines, and large-scale dataset analysis.
- 4 years of experience in algorithms and computation, including data structures, algorithm design, and computational optimization.
- 4 years of experience in programming and scripting languages, including Python, R, Java, JavaScript, PHP, Angular, and Docker.
- Strong experience in enterprise data management and system architecture, designing scalable database solutions and data-driven applications for research and business intelligence.
- Extensive experience in feature selection, model evaluation, and automation of data-driven processes, ensuring high-performance analytical workflows.

DATABASES:

MySQL 5.7/8.0 (SQL, Stored Procedures, Query Optimization); PostgreSQL 11/13 (Indexing, Partitioning, Performance Tuning); SQL Server 2016/2019 (T-SQL, Database Administration, Backup & Restore Security Policies)

MACHINE LEARNING LIBRARIES:

Scikit-learn (Classification, Regression, Clustering, Feature Engineering); XGBoost (Gradient Boosting, Hyperparameter Tuning, Model Optimization); bnlearn (Bayesian Networks, Probabilistic Graphical Models); e1071 (Support Vector Machines, Naïve Bayes); randomForest (Ensemble Learning, Feature Selection, Model Interpretability);

WEB DEVELOPMENT & APIs:

PHP (Backend Development, Secure Authentication, API Development); Angular (Frontend Development, Component-Based Architecture, API Integration); REST APIs (Data Exchange, Authentication Mechanisms, Security Best Practices)

CLOUD & CONTAINERIZATION:

Docker (Container Orchestration, Application Deployment, Database Management in Containers); Virtual Machines (Cloud-Based Deployment, Server Management, Database Hosting)

SOFTWARE:

MySQL Workbench, PostgreSQL pgAdmin, Microsoft SQL Server Management Studio (SSMS), Jupyter Notebook, Visual Studio Code, Docker, Apache Airflow, Tableau, Power BI, GitHub, Jenkins, Postman (API Testing)

OPERATING SYSTEMS:

Windows Server 2016/2019

HARDWARE:

PCs (Intel x86/x64, AMD Ryzen), Server Architectures (Cloud Virtualized, Dedicated Hosting), Storage Devices (SSD, NVMe, RAID Configurations)

WORK EXPERIENCE:**Educational Technology Coordinator – Little Neighborhood School Inc., Sherman Oaks, CA:**

Duration: 2017 – 2023

- Designed and maintained an SQL database to manage student, staff, and administrative records, including enrollment, billing, and attendance.
- Performed regular database backups, integrity checks, and query optimization to enhance system reliability and security.
- Assisted in implementing and troubleshooting educational software, digital learning tools, and interactive platforms to improve classroom engagement.
- Provided technical support for staff and parents, including training on digital communication platforms and school portals.
- Developed custom SQL queries and reports for student progress tracking, billing summaries, and administrative analytics.
- Maintained the school's website, ensuring accurate program, schedule, and event updates.
- Designed digital flyers, newsletters, and interactive content using tools like Canva to improve communication with parents.

Technologies Used:

- SQL (MySQL), Microsoft Office Suite (Excel, Word, PowerPoint), Google Workspace, Canva

Predicting Housing Tenure Decisions in Los Angeles Using Machine Learning Project, CA

Duration: 2024 January - 2024 December

- Investigated key factors influencing homeownership vs. renting in Los Angeles using machine learning.
- Compiled and preprocessed American Housing Survey data (2015–2021), performing feature engineering and multivariate imputation for missing values.
- Implemented Random Forest, XGBoost, Gradient Boosting Machines, SVM, KNN, and Naïve Bayes models, achieving a 99.93% accuracy with Random Forest.
- Utilized AdaBoost for feature ranking, identifying 125 key variables from an initial 3,000+ predictors.
- Evaluated models using 10-fold cross-validation and metrics such as accuracy, precision, recall, and confusion matrix analysis.

Key Findings:

- Identified building type, household income, and utility costs as the most influential factors in tenure decisions.
- Contributed to a deeper understanding of housing market dynamics in Los Angeles.

Technologies Used:

- R (caret, xgboost, gbm, train), Python, MySQL, American Housing Survey datasets, RStudio

Data Mining Analyst – "The Crime in California's Cities" Project, Northridge, CA

Duration: February 2024 – May 2024

- Conducted an in-depth analysis of murder rates in California cities using FBI crime data (2005–2019).
- Designed and deployed a web scraper to automate data extraction from the FBI's website.
- Performed data cleaning, integration, and feature engineering using R libraries (tidyverse, dplyr, caret).
- Applied Linear Regression and Random Forest models for predictive analytics, leveraging 5-fold cross-validation and evaluation metrics (RMSE, MAE, MAPE).
- Conducted data visualization using histograms, scatterplots, and correlation matrices for feature selection.
- Transformed regression outputs into classification ranges for Confusion Matrix evaluation, improving model interpretability.
- Published cleaned datasets, scripts, and results on GitHub for reproducibility and collaboration.

Key Achievements:

- Developed a data-driven model for crime prediction and analysis.
- Automated data collection and preprocessing, reducing manual effort.

Utilized Technologies:

- R (tidyverse, caret, ggplot2), Python, GitHub, web scraping tools, statistical analysis frameworks

Database Engineering Project – California State University, Northridge, CA

Duration: September 2024 – December 2024

- Led backend development and database architecture for a rental management application.
- Designed and implemented normalized database schemas, ensuring data integrity and efficiency.
- Developed secure RESTful APIs using PHP for user authentication, rental listing creation, search functionalities, and review management.
- Enforced SQL security best practices, including prepared statements, input validation, and password hashing.
- Integrated the database system with an Angular frontend, ensuring a seamless user experience.
- Deployed the platform using Docker, allowing for scalable and efficient application maintenance.

Key Contributions:

- Developed a functional rental property listing and review system with enforced business rules.
- Ensured data security and integrity through SQL injection prevention and access controls.

Utilized Technologies:

MySQL, PHP, Git, Visual Studio Code, Docker, MySQL Workbench 8.0 CE

Math Tutor - University of California, Los Angeles (UCLA), CA:

Duration: 2019 – 2021

- Provided one-on-one tutoring in geometry, precalculus, and algebra for undergraduate students.
- Created custom lesson plans and practice problems tailored to individual learning needs.
- Supported first-generation college students and English language learners to improve academic performance.

Caregiver – Premier Healthcare Services Inc., Los Angeles, CA:

Duration: February 2018 – 2024

- Provided personalized care for individuals with special needs, ensuring safety, well-being, and daily support.

- Collaborated with families and healthcare professionals to implement effective care plans.
- Assisted with daily activities, therapy sessions, and personal development routines.

Preschool Teacher – My Lastochka School, CA:

Duration: 2013 – 2024

- Developed lesson plans and interactive learning activities for preschool-aged children.
- Provided language development support in a multilingual classroom environment.

Skills

- Programming Languages: R, Python, Java, C++, SQL, C#, PHP
- Machine Learning: Supervise/Unsupervised Learning, Feature Engineering,
- Data Mining: Extract, Transform, Load operations
- MySQL Database: Normalization, Query Optimization, Indexing, Transactions
- MS SQL Server: T-SQL, Stored Procedures, Query Plan Analysis, Performance Tuning
- Algorithm Design and Optimization: Time Complexity Analysis, Greedy & Dynamic Programming)
- Software Engineering Practices: Agile, Scrum, CI/CD Pipelines
- Data Visualization (Matplotlib, Seaborn, ggplot2)
- Trilingual
- Strong Problem-Solving & Critical Thinking in Data Science & Engineering
- Excellent Communication & Team Collaboration

EDUCATION:

- Master of Science in Computer Science - California State University, Northridge, USA
- B.S. in Mathematics - University of California, Los Angeles, USA

ADDITIONAL INFO:

- US work authorization: Active
- Other: trilingual, non-smoker, active in sports, fast and curious learner
- References: Available upon request